

**A**

## Genomes

- Local assemblies
- NCBI accessions

## Reference loci

>Af293|*benA*  
 ATGCGTGAAATCGTATGTTGTGC...  
 >Af293|*rpb2*  
 CAGGATCCATCAGCAACCGGAG...

## Configuration

- Loci of interest
- Tree method + bootstrap
- ANI method

**B**

## Locus mapping & extraction

- Mapping & filtering
- Hit extraction & joining
- Locus classification

| A1163      | <i>benA</i> |
|------------|-------------|
| Position   | DS499594.1  |
| Similarity | 100%        |
| Coverage   | 100%        |
| Length     | 1983 bp     |

| Sample   | <i>actin</i> | <i>benA</i> | <i>calmodulin</i> | <i>rpb2</i> |
|----------|--------------|-------------|-------------------|-------------|
| A1163    | 1            | 1           | 1                 | 1           |
| Af293    | 1            | 1           | 1                 | 1           |
| IFM46972 | 1            | 1           | 1                 | 1           |
| IFM46973 | 1            | 1           | 1                 | 1           |
| IFM58399 | 1            | 1           | 1                 | 1           |
| NRRL181  | 1            | 1           | 1                 | 1           |
| NRRL4585 | 1            | 1           | 1                 | 1           |
| PK20-01  | 1            | 1           | 1                 | 1           |

**C**

## Pairwise ANI calculation

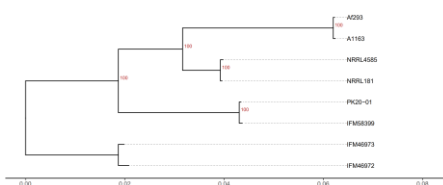
skani / fastANI / pyANI

|          | Af293 | NRRL181 | A1163 | IFM46973 | IFM46972 | IFM58399 | NRRL4585 | PK20-01 |
|----------|-------|---------|-------|----------|----------|----------|----------|---------|
| Af293    | 100   | 91.28   | 99.59 | 87.86    | 87.82    | 89.65    | 91.73    | 89.55   |
| NRRL181  | 91.28 | 100     | 91.4  | 89.96    | 89.93    | 91.7     | 99.13    | 91.64   |
| A1163    | 99.59 | 91.4    | 100   | 87.96    | 87.81    | 89.64    | 91.73    | 89.57   |
| IFM46973 | 87.86 | 89.96   | 87.96 | 100      | 98.97    | 89.82    | 90.05    | 89.8    |
| IFM46972 | 87.82 | 89.93   | 87.81 | 98.97    | 100      | 89.78    | 89.87    | 89.65   |
| IFM58399 | 89.65 | 91.7    | 89.64 | 89.82    | 89.78    | 100      | 92.15    | 99.32   |
| NRRL4585 | 91.73 | 99.13   | 91.73 | 90.05    | 89.87    | 92.15    | 100      | 92.13   |
| PK20-01  | 89.55 | 91.64   | 89.57 | 89.8     | 89.65    | 99.32    | 92.13    | 100     |

**D**

## MLSA phylogeny

- Per-locus alignment MUSCLE
- Concatenate + partition
- IQ-TREE / RAxML / FastTree

**E**

## Results

- QC Reports
- Detailed tool logs
- Extracted loci of interest
- MLSA tree + ANI heatmap
- Automated Snakemake report

